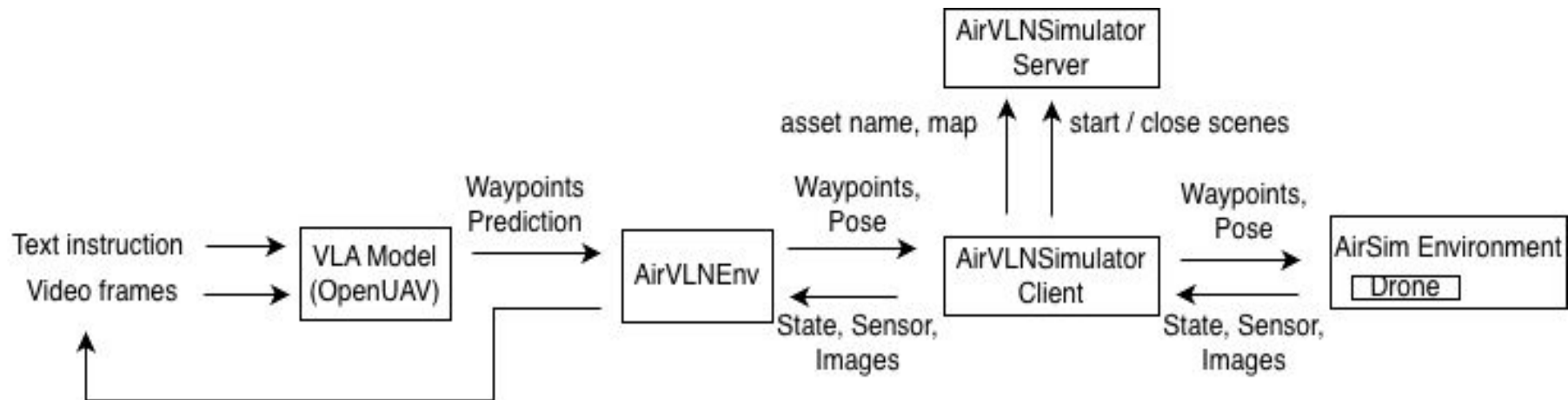


Aerial Vision-Language Navigation on OpenUAV

Research Objective and Sub-goals

- Train and evaluate a Vision-Language Action (VLA) model
- Expand the model's **capability horizon**
 - re-evaluation/comparison with original baseline results
- Analyze the datasets (input/output)
- Analyze the **weakness and strengths** of the OpenUAV model and **benchmarks**

System Overview



An Example of Instruction Format

```
hoon > TravelUAV > simulator_environments > NYCEnvironment
1  ✓  {
2    "object_name": "SM_Woman53",
3    "start": [
4      -308.166,
5      -130.412,
6      -7.8919999999999995
7    ],
8    "end": [
9      -278.3994140625,
10     -251.08233642578125,
11     -4.858692646026611
12   ],
13   "target": {
14     "position": [
15       -278.3994140625,
16       -251.08233642578125,
17       -0.8586926460266113
18     ],
19     "rotation": [
20       0,
21       0
22     ]
23   }
24 }

"trajectory_raw_detailed": [
{
  "position": [
    -308.1659851074219,
    -130.41200256347656,
    -7.892000198364258
  ],
  "orientation": [
    0.0,
    0.0,
    0.8068441152572632,
    -0.5907643437385559
  ]
},
{
  "position": [
    -308.1177673339844,
    -130.5727996826172,
    -8.554638862609863
  ],
  "orientation": [
    0.006659571547061205,
    0.007993844337761402,
    0.5552759170532227,
    -0.831601083278656
  ]
},
{
  "position": [
    -308.011962890625,
    -130.92530822753906,
    -9.362601280212402
  ],
  "orientation": [
    0.0009209764539264143,
    0.0012315195053815842,
    0.603247880935669,
    -0.7975523471832275
  ]
}
]

{
  "trajectory": [ ...
  ],
  "trajectory_raw": [ ...
  ],
  "trajectory_raw_detailed": [ ...
  ],
  "image_feature_path": "/home/hoon/TravelUAV/simulator_environments/NYCEnvironmentMegapa/0a3caa7e-947e-4c07-b31b-b...",
  "index": [ ...
  ],
  "length": 35,
  "conversations": [
    {
      "from": "human",
      "value": "<image>\nThere is a target in the right front of uav. Using your front as the x-axis and your ..."
    },
    {
      "from": "gpt",
      "value": ""
    }
  ]
}
```

My Accomplishments

- OpenUAV setup and configuration
- Formatting results extraction
- Video generation for each scene
- 3D trajectory plotting
- Process ID force-termination script

Qualitative Results

Assistant Mode	Scene ID	Instruction	Target	Result	GT Path Length	Taken Path Length	Navigation Error (m)	Success Path Length (%)
L1	2569c9f7-b353-424b-8369-295c343215c6	There is a target in the left front of uav. Using your front as the x-axis and your right as the y-axis, The target is at a yaw angle of -64.0 degrees from you. A yellow car is parked on a dark asphalt road surrounded by high-rise buildings with trees and sidewalks, located at a street intersection. Please control the drone and find the target.	yellow car	Fail	477.97	486.76	33.13	0
L1	49eb5067-7c93-487a-a67e-e78488eacfe1	There is a target in the left front of uav. Using your front as the x-axis and your right as the y-axis, The target is at a yaw angle of -65.0 degrees from you. The red car is located on a city street with multi-story buildings lining both sides, visible under clear daylight; there is a street intersection nearby, and the surroundings include a telephone booth and a few lamp posts. Please control the drone and find the target.	red car	Fail	416.86	429.71	77.88	0
L1	04c4ffd7-b7cf-47e8-bc63-86b445399490	There is a target in the back of uav. Using your front as the x-axis and your right as the y-axis, The target is at a yaw angle of -177.0 degrees from you. The red motorcycle is parked on a light-colored pavement with shadows, near a waterfront with benches and a bridge in the background, surrounded by trees and adjacent to multi-story buildings and lamp posts lining a straight road. Please control the drone and find the target.	red motorcycle	Oracle Success	115.10	159.12	14.4	72.34
L1	8e029be9-e4c6-4f45-bfd6-63235ae3a520	There is a target in the back of uav. Using your front as the x-axis and your right as the y-axis, The target is at a yaw angle of 172.0 degrees from you. The black car has a reflective surface, a sleek design, and is parked on a bridge with suspension cables visible in different angles. The bridge is surrounded by a large body of water, and a cityscape with tall buildings is visible in the distance, under a clear sky. Please control the drone and find the target.	black car	Success	149.36	162.99	13.22	91.64
L1	941702fd-5f21-4e80-98f3-9a8e623af023	There is a target in the right of uav. Using your front as the x-axis and your right as the y-axis, The target is at a yaw angle of 78.0 degrees from you. A man wearing black and white clothing is standing on a wide curved path made of pavement with a shadow on the ground. The surroundings feature tall buildings with multiple windows on one side of the path, a bridge visible in the distance, a few trees, streetlights, and seating areas near a waterfront to the right with a green signal light overhead. Please control the drone and find the target.	man wearing black and white clothing	Success	115.07	187.72	9.72	61.3

Video generated out of frames

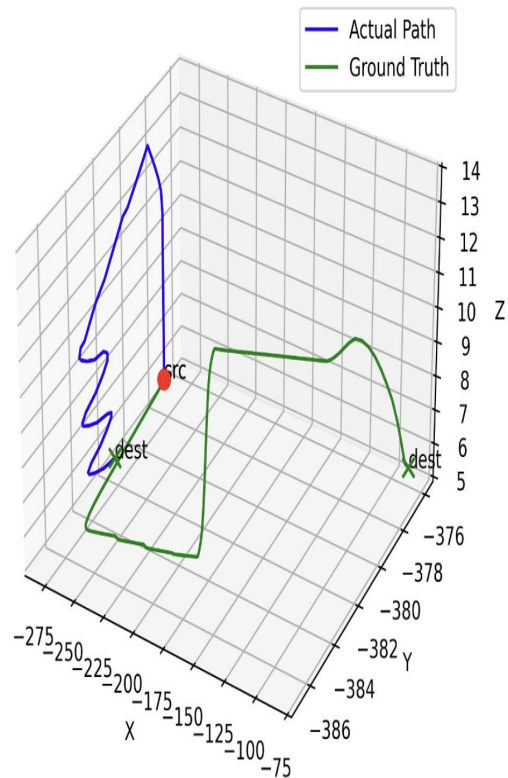
["A man wearing a white shirt is located on a wide paved street near a curve, flanked by tall buildings on one side and an open area with trees and views of a bridge and body of water in the background. The surrounding environment features urban elements like street signs, sidewalks, and a waterfront area with benches and a distant cityscape."]



3D Trajectory Plots

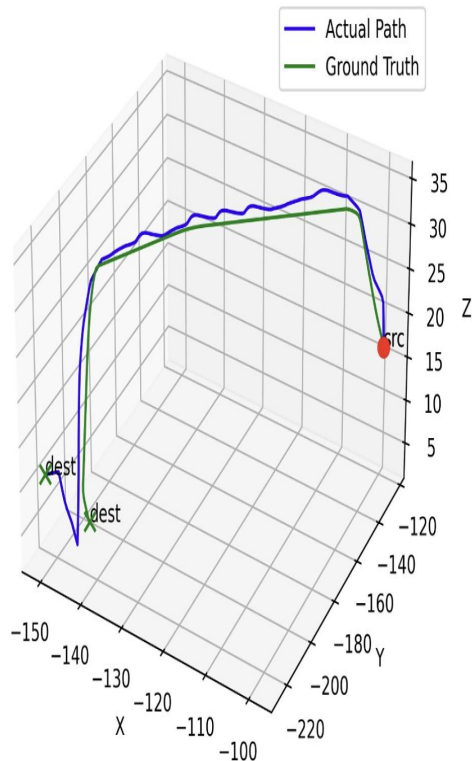
Fail

Traj: 4c8d7356-7659-4441-97e4-64a217346788



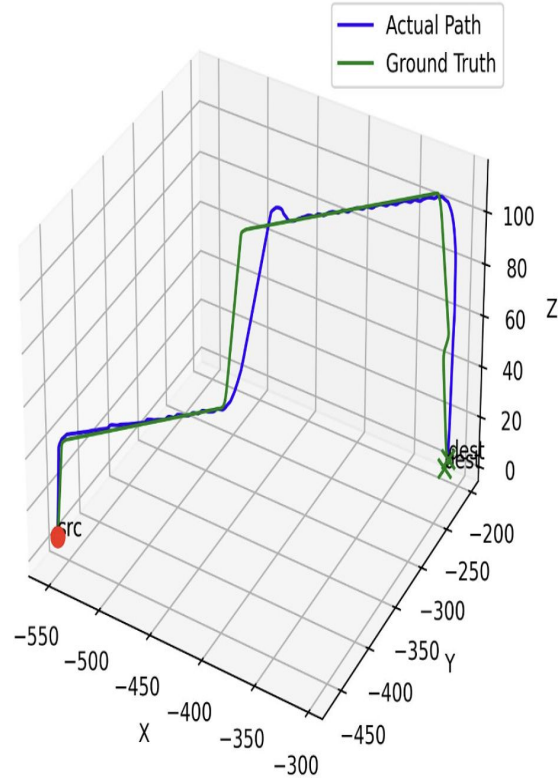
Oracle Success

Traj: oracle_04c4ffd7-b7cf-47e8-bc63-86b445399490



Success

Traj: success_b23e79a3-4f1d-43e4-87f6-d9ffe866cb24



Physical Drone Deployment

